

What are Object Creational Patterns?

The first part of this 12-part series of articles on software design focuses on 'creational patterns'. The simple words and examples used here will help newbies understand these patterns better.



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The wheel is said to be one of man's greatest inventions. However, reinventing the same wheel again doesn't make any sense. A better approach is to come up with a new idea using the wheel. The same logic holds good for software design as well.

Most of the design problems that we attempt to solve might have been solved already by someone else. It makes more sense to use those known design ideas to solve a bigger and a newer problem than breaking heads in reinventing the same idea again.

The famous Gang of Four (Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides) published a good number of reusable design ideas way back in the mid-90s, in a book titled 'Design Patterns: Elements of Reusable Object-Oriented Software'. As the title suggests, they called the design ideas 'design patterns' and the book focused on the area of object-oriented design.

Since then, several experts have published hundreds of design patterns applicable to various other design areas like domain driven design, enterprise integration, distributed computing, microservices, system architecture, transaction management, data management, etc.

Though there is a lot of literature available on these design patterns, it is full of jargon. Because of this, many greenhorns find it really difficult to understand and apply them. This 12-part series attempts to present a few of the patterns in simple words with the help of illustrations and case studies.

Providers and consumers

Let us build a user management system (UMS) that maintains a list of users by their phone numbers. Does it look trivial? Of course, yes! However, this system is good enough for us to understand a couple of design patterns. Figure 1 presents a basic design for the UMS.

The package *com.glarimy.ums* has a class *InMemoryUserRepository* that holds the user entries in a map. The phone number of the user acts as the *key* while the name

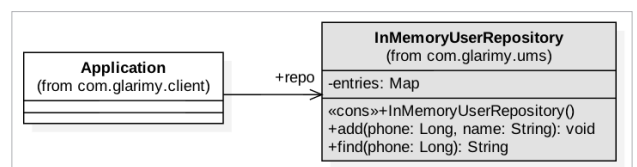


Figure 1: Basic design for a user management system